AI-Assisted Legacy Modernization

Pilot report and proposal: a COBOL mainframe application translated to Rust with Claude Code

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In one slide

What we did

- Took a complete mainframe application: a credit-card system in **COBOL**, with online screens, nightly batch jobs and mainframe data files
- Had an AI coding agent, **Claude Code**, translate it into **Rust**, a modern, safe language
- Required **proof**: old and new programs run on the same data and must give the same output

What we got

- All **30 programs** translated, working and tested
- **40 of 40** batch outputs and **338 of 338** screens identical to the original
- **42 of 42** test cases passed
- **15 defects** found in the original and documented
- About **3 hours** of AI working time, one engineer supervising

Recommendation: run a second pilot on one of our own legacy systems, with the same method.

Why legacy code is a business risk

The situation

- Core business rules live in decades-old COBOL, PL/I or RPG code
- The people who wrote it are retiring; few new engineers learn these languages
- Mainframe capacity and licences are a large, fixed cost
- Manual rewrites are slow and risky: rules are buried in the code, and small differences in rounding, dates or edge cases break the business

What a good solution must do

1. Understand the old code, *including its quirks*
2. Produce maintainable code in a modern language
3. **Prove** the new system behaves like the old one
4. Keep people in control of scope, review and release
5. Be fast and repeatable across many systems

The original application: what CardDemo is for

CardDemo is the core system of a **credit-card issuer**, the kind a bank runs on its mainframe: who the customers are, which cards and accounts they hold, what they spent and what they owe.

Who uses it

- **Bank staff**, through green-screen terminals: accounts, cards, transactions, bill payments, reports
- **Administrators**: which staff may sign on, and with which role
- **The nightly batch**: unattended programs that process the day's card activity

What it holds

- Customers, their accounts and credit cards
- Balances, credit and cash limits, billing-cycle totals
- Every card transaction: purchases, returns, payments, interest
- Transaction types, categories, interest-rate groups
- Staff user ids and roles

A day in the life of the system

During the day: online screens

- **Sign on**: menu by role
- **Accounts**: view, change limits and address
- **Cards**: list, view, block, edit
- **Transactions**: browse, view, add
- **Bill payment**: pay a balance in full
- **Reports**: request for any period
- **Users**: add, change, delete staff

At night: the batch cycle

1. **Posting**: check, post or reject the day’s transactions
2. **Interest**: per account and category
3. **Combine** daily and interest transactions
4. **Statements**: per customer, text and web
5. **Reports**: page, account and grand totals
6. **Export / import** for another branch

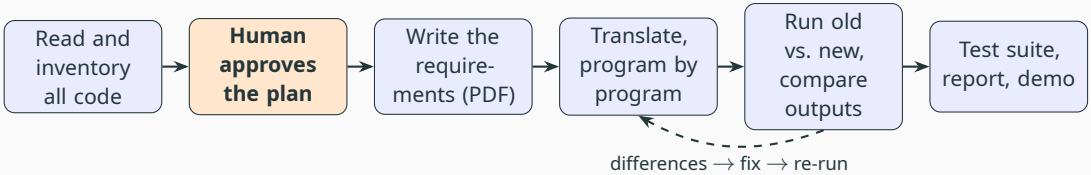
What was translated

	Original (COBOL, mainframe)	Translation (Rust)
Online screens	17 programs (IBM CICS), 17 screen layouts	17 modules + terminal screen engine
Nightly batch	13 programs + JCL job scripts	13 modules + job runner
Data	VSAM files, EBCDIC encoding, packed decimals	Embedded database, same record layouts and bytes
Size	≈ 27,000 lines	≈ 10,800 lines (incl. generated code)

The application is **AWS CardDemo**, a public sample built to represent real mainframe banking systems. The original code is unchanged in the repository (`cobol/`), next to the translation (`rust/`).

CardDemo © Amazon.com, Inc. or its affiliates, used under the Apache License 2.0. The Rust translation is a derived work by Illia Burov. This pilot is not affiliated with or endorsed by Amazon Web Services.

How the AI worked: plan first, prove at the end



- Before writing code, the AI asked what to include. A person chose the scope: base application, terminal screens, embedded database, the original's behaviour kept exactly.
- Every phase ended in a saved, reviewable step (25 commits), with a written log of decisions.
- The AI also produced the documents people need: requirements, test suite, demo guide, this report.

Results at a glance

Measured results

Measure	Result
Programs translated	30 / 30
Batch outputs vs. COBOL	40 / 40 identical
Screens vs. COBOL	338 / 338 identical
Manual test cases	42 / 42 passed
Defects found in the original	15
AI working time	≈ 3.1 h

Documents produced by the AI

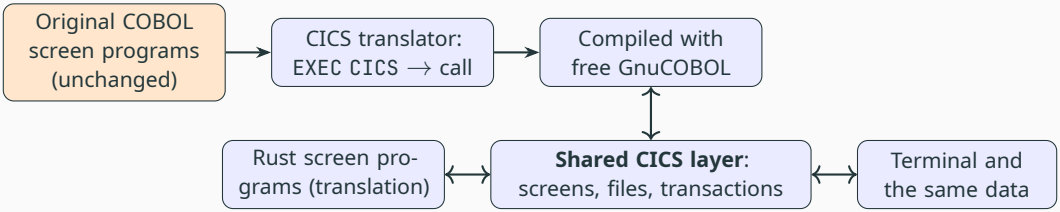
- Requirements specification: 26 pages, 165 functional requirements
- Manual test suite with execution log
- Live demo guide for non-technical audiences
- Progress report with time and token figures
- `CLAUDE.md`: the rulebook for future AI work on this code

Proof: the old and new systems side by side

Nightly job	What is compared	Result
Daily posting	updated accounts, balances, transactions, rejected items	✓ identical
Interest calculation	accounts, generated interest transactions	✓ identical
Transaction report	519-line printed report	✓ identical
Customer statements	50 statements, text and web version (7,900 lines)	✓ identical
Data export / import	migration file, re-imported files, error file	✓ identical
File print utilities	5 programs, incl. packed-decimal outputs	✓ identical
Online screens	338 screens of 49 scripted terminal sessions: every field, colour, cursor	✓ identical

Original COBOL (free compiler) and Rust run on **the same data, clock and keystrokes**; one command each repeats the checks.

The original online screens, live



- Real CICS runs only on licensed mainframes: the original programs get a **stand-in for CICS**, the layer the Rust version already uses. Both runs share it, so the comparison checks the business logic
- All 17 original screen programs run **unchanged**; old and new screens can be shown **side by side**
- Automated comparison: **338 of 338 screens identical**; on the way it found 13 differences in the Rust port (all fixed) and 3 more defects of the original (documented)

Why not run it on an emulated mainframe?

The idea

Hercules, a free PC emulator of IBM mainframe hardware: why not run the original on it?

What a free setup can run

- MVS 3.8j, the operating system from 1981 that may be used freely
- No CICS: only the hobbyist look-alike **KICKS**
- Only a 1970s COBOL compiler (ANSI COBOL 74)

Why that does not work for CardDemo

- CardDemo is modern **Enterprise COBOL** (COBOL 85 and later: EVALUATE, END-IF, inline PERFORM, intrinsic functions); it does not compile with a COBOL 74 compiler
- Making it compile would mean **rewriting the original**, the opposite of what the demo must show
- Current z/OS with real CICS on a PC needs IBM's **licensed** test environment (ZD&T)

The CICS bridge runs the original code unchanged, at no licence cost.

What the AI found that people had missed

In the original system

Kept and documented as *known defects*:

- Last account's interest is never saved
- The report counts the last amount twice
- Card update wipes the security code
- Account update moves the group id
- Two programs do not compile
- … 15 in total

In its own translation

Found by running the original, and fixed:

- Batch comparison: 9 subtle differences
- AI **code review** (scoped by `CLAUDE.md`): 10 issues
- Online comparison: 13 differences
- All fixed; none was visible in normal tests

The value is not only in writing code fast, but in verifying it automatically.

CLAUDE.md: the project rulebook for the AI

What it is

A plain-text file in the repository, read by Claude Code at the start of **every** session:

- **Scope**: what may change
- **How to build, test, prove**
- **Review policy**
- **Lessons learned**

Excerpt

```
## Code review
Review only rust/.
- cobol/ is the original source and
  the specification: never report
  findings against it.
- A difference from the COBOL is a
  Rust bug. Known defects (KD-nn)
  are kept on purpose.
```

One reviewed, versioned rulebook keeps every session consistent.

Human oversight built into the workflow

Control points

- **Plan approval**: no code until a person accepted plan and scope
- **Checkpoints**: every step is a commit that can be reviewed or undone; nothing is published unasked
- **Budget limits**: the AI stopped at 85% of the usage quota and wrote a hand-over note; work resumed exactly there the next day
- **Memory and logs**: decisions and findings are written down

Claude Code Remote Control

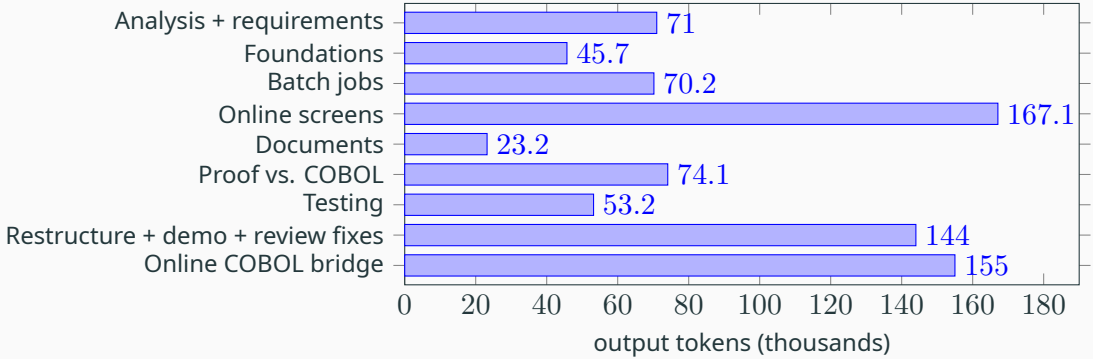
- The work runs on the company workstation; the code stays there
- From the **Claude mobile app**, the engineer follows the session, answers the AI's questions and approves steps while away from the desk
- Long runs no longer need someone sitting at the screen

Time and tokens

Session	Work	AI time	Output tokens
1	Analysis, requirements, translation of all 30 programs, test suite, report	1.1 h	367k
2	Proof against the original COBOL, test execution, fixes	0.5 h	137k
3	Restructure, demo tooling and guide, `CLAUDE.md`, code-review fixes, this presentation	0.9 h	144k
4	Original COBOL screens on a CICS stand-in, online comparison, fixes	0.6 h	155k
Total		≈ **3.1 h**	≈ **0.8M**

- Measured from the Claude Code session logs (pauses over 10 minutes removed).
- Token cost depends on the subscription or API price list in use.
- The engineer’s own time went into decisions and reviews, not into typing code.

Where the effort went



The online screens were the largest part: 17 interactive programs with many validation rules each.

Risks and mitigations

Risk	Mitigation in this method
AI produces plausible but wrong code	Run old and new on the same data; outputs must be identical
Hidden rules in old code are lost	Requirements are written first, from the code, and reviewed
Original defects get “fixed” silently	Defects are kept and documented; fixing them is a business decision
Parts that cannot run off the mainframe (online CICS screens)	Screens drawn from the original layouts; 41-case test suite; next: comparison on a test mainframe
Source code confidentiality	Code stays on company machines; only what is sent to the model leaves them, under the company’s agreement with the vendor
Over-reliance on one tool	Output is ordinary Rust code plus documents; any team can maintain it

From pilot to programme

Next pilot (4–6 weeks)

1. Pick one of our legacy systems: bounded, with test data
2. Write its `CLAUDE.md`: scope, target stack, review rules
3. Set up the comparison harness: old vs. new on the same data
4. Translate, verify, document, as in this pilot
5. Review with the system owners; agree go-live criteria

What we need

- One engineer who knows the system, part time
- Access to the code and representative test data
- Claude Code licences for the team
- A test environment for the original system

Success criteria

Identical outputs on agreed test data, test suite passed, owners sign off.

Live demo

```
python3 demo/demo.py cobol      # the original COBOL programs
python3 demo/demo.py rust       # the Rust translation
python3 demo/demo.py compare    # → IDENTICAL, file by file
python3 demo/demo.py online-cobol    # original screens; online = Rust
python3 demo/demo.py online-compare  # → 338/338 screens identical
```

Step-by-step guide for presenters and non-technical audiences: `docs/DEMO.md`.

Questions: Illia Burov

Appendix

Technical architecture

Component	Replaces	Lines
`cobol`	COBOL data rules: decimals, EBCDIC; records generated from copybooks	1,677
`vsam`	VSAM indexed files, on SQLite	556
`cics`	CICS screens (from the original BMS files), transfers, file control	833
`online`	the 17 online programs, one function per COBOL paragraph	4,283
`batch`	the 13 batch programs and JCL jobs	1,886
`carddemo`	the application: terminal screens, job runner	440

Verification: `rust/difftest/run.py` (GnuCOBOL comparison) and `cargo test` (21 suites, including the 41 manual cases as scripted terminal sessions and the multi-user tests).

Known defects of the original (selection)

ID	Where	Effect
KD-01	Interest calculation	the last account’s interest is never written back
KD-02/03	Transaction report	last amount counted twice; no final account total
KD-07	Card update screen	the card security code is wiped
KD-12	Export / import	the programs do not compile as shipped
KD-13	Account update screen	the group id is saved into the ZIP-code field
KD-14	Admin menu	after a not-installed option, the next input is ignored
KD-15	Account update screen	an account with a blank group never counts as unchanged

All 15 are in the requirements specification, with the exact behaviour the translation reproduces.